Proofs of the conjectured column and row recurrences for OEIS A275266

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1 The conjectures

The Online Encyclopedia of Integer Sequences records A275266 as

> *T(n,k)=Number of nXk 0..2 arrays with no element equal to any value at offset (-2,-1) (-2,1) or (-1,-1) and new values introduced in order 0..2*

a table read by upward antidiagonals, and carries in its Formula section the following empirical lines, none of them proved there:

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Empirical for column k:
k=1:  a(n) = 4*a(n-1) -3*a(n-2)
k=2:  [order 9] for n>10
k=3:  [order 14] for n>17
k=4:  [order 43] for n>47

Empirical for row n:
n=1:  a(n) = 4*a(n-1) -3*a(n-2)
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The lines carry no separate signature; the entry itself is by R. H. Hardin (Jul 21 2016). As of revision 4, last modified Jul 21 2016, the entry records no proof of any of them and links to none. This note proves the 4 column recurrences and the 1 row recurrence listed above.

The entry also carries further lines of the same blocks that this note does not settle; they are named in Section 6.

2 The object, and what a column and a row are

The array condition is the same for every width; it is written out here for width 4, which is one of the widths this note uses.

Let A be an $n' \times 4$ array over $\{0, \dots, 2\}$ with $n' = n$ rows.

The entry's offsets are (-2,-1), (-2,1), (-1,-1). The condition is

$$A[i][j] \neq A[i+\delta][j+\varepsilon] \quad \text{for every listed } (\delta, \varepsilon) \text{ landing inside the array.}$$

An offset and its negative say the same thing, so the offsets are normalised here to point upwards or, in the same row, leftwards: (-2,-1), (-2,1), (-1,-1).

The entry's further clause, that the new values $0, \dots, 2$ are introduced in order, means that reading the array in row-major order the first occurrences of the values happen in increasing order: a value v may appear only once every smaller value has already appeared.

$T(n,k)$ is the number of such arrays with $n+0$ rows and $k+0$ columns. Column $k=K$ is therefore the sequence $n \mapsto T(n,K)$, an array count of fixed width $K+0$; row $n=N$ is the sequence $k \mapsto T(N,k)$, which is the same count with the roles of the two directions exchanged — an array count of fixed width $N+0$ read across.

The entry publishes its table by upward antidiagonals, so every published term is a value $T(n,k)$ with both indices known. All 46 of them are used as the data gate below: the models are required to reproduce the whole published triangle, not merely the column or row a given line is about.

3 Each column and each row is a walk count

Fix the width. The condition on a cell reaches at most one row up and one row down, so a window of three consecutive rows decides the whole of its middle row, and an array of n rows is exactly a walk of length $n-1$ in the digraph whose states are pairs of consecutive rows, starting at a state with no row above and ending at an accepting one. With M its adjacency matrix, w the indicator of the starting states and f of the accepting ones, the count is $w^{\mathsf{T}} M^{n-1} f$, hence C-finite.

States from which the same number of arrays can be completed, for every remaining number of rows, are identified by partition refinement: begin with the partition by acceptance and separate two states as soon as they send different numbers of edges into some block. On the blocks of the result the number of completions of each length is constant, so the quotient counts exactly what the original does; the mirror refinement, on the edges coming in rather than going out, is then applied in the same way. Writing L for the resulting matrix and S for its size, the sequence satisfies the linear recurrence given by the characteristic polynomial of L, of degree S. That is the degree bound used below, and it is computed separately for every line, never assumed.

For a claimed recurrence with characteristic polynomial q, the residual $u_m = \tilde{w}^{\mathsf{T}} L^{m-1} q(L) \tilde{f}$ is exactly the residual of the claim at one index and itself satisfies the characteristic polynomial of L. So S consecutive vanishing residuals force every later one, the test is finite, and the last nonvanishing residual pins the threshold exactly.

4 Results

Theorem 1. *For column $k=1$ of A275266, the recurrence $a(n) = 4\,a(n-1) - 3\,a(n-2)$ holds for every $n \geq 3$, at every index at which it can be stated.*

Proof. The width is 1; the digraph has 16 states, 16 after trimming, and $S=3$ blocks after refinement. The model reproduces every published value of column $k=1$ — and of the rest of the triangle. The residuals were computed exactly and all of them vanish; after that point more than $S=3$ consecutive residuals vanish, so by Section 3 every later one vanishes too. □

Theorem 2. *For column $k = 2$ of A275266, the recurrence $a(n) = 0$ holds for every $n > 10$, and fails at $n = 10$.*

Proof. The width is 2; the digraph has 61 states, 61 after trimming, and $S = 11$ blocks after refinement. The model reproduces every published value of column $k = 2$ — and of the rest of the triangle. The residuals were computed exactly and the last nonvanishing one is at $n = 10$; after that point more than $S = 11$ consecutive residuals vanish, so by Section 3 every later one vanishes too. ☐

The entry states this line for $n > 10$; the proved threshold is $n > 10$.

Theorem 3. *For column $k = 3$ of A275266, the recurrence $a(n) = 0$ holds for every $n > 17$, and fails at $n = 17$.*

Proof. The width is 3; the digraph has 278 states, 278 after trimming, and $S = 25$ blocks after refinement. The model reproduces every published value of column $k = 3$ — and of the rest of the triangle. The residuals were computed exactly and the last nonvanishing one is at $n = 17$; after that point more than $S = 25$ consecutive residuals vanish, so by Section 3 every later one vanishes too. ☐

The entry states this line for $n > 17$; the proved threshold is $n > 17$.

Theorem 4. *For column $k = 4$ of A275266, the recurrence $a(n) = 0$ holds for every $n > 47$, and fails at $n = 47$.*

Proof. The width is 4; the digraph has 1063 states, 1063 after trimming, and $S = 71$ blocks after refinement. The model reproduces every published value of column $k = 4$ — and of the rest of the triangle. The residuals were computed exactly and the last nonvanishing one is at $n = 47$; after that point more than $S = 71$ consecutive residuals vanish, so by Section 3 every later one vanishes too. ☐

The entry states this line for $n > 47$; the proved threshold is $n > 47$.

Theorem 5. *For row $n = 1$ of A275266, the recurrence $a(k) = 4\,a(n-1) - 3\,a(n-2)$ holds for every $k \geq 3$, at every index at which it can be stated.*

Proof. The width is 1; the digraph has 16 states, 16 after trimming, and $S = 3$ blocks after refinement. The model reproduces every published value of row $n = 1$ — and of the rest of the triangle. The residuals were computed exactly and all of them vanish; after that point more than $S = 3$ consecutive residuals vanish, so by Section 3 every later one vanishes too. ☐

5 What is not settled here

The same blocks carry further lines that this note leaves open, with the reason for each:

- 3 lines: model does not reproduce the published table
- 3 lines: fewer than six published terms in that line

6 Verification

- **The published table.** Every one of the 46 values $T(n,k)$ the entry publishes was recomputed from the models and agrees exactly, in integer arithmetic, with the indices read off the entry's offset (1) and its stated shape.
- **The claims on the raw data.** Each recurrence was evaluated on the entry's published values alone, with no model involved, wherever the proved range and the published data overlap.
- **The annihilation test.** Carried to the derived bound for each line separately, over the whole vector, in exact integer arithmetic.

7 References

1. OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A275266, revision 4, last modified Jul 21 2016.
2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7.
3. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011.